X(3872) Analysis Update

PbPb $\psi(2S)$ validation and the PbPb23 X extension

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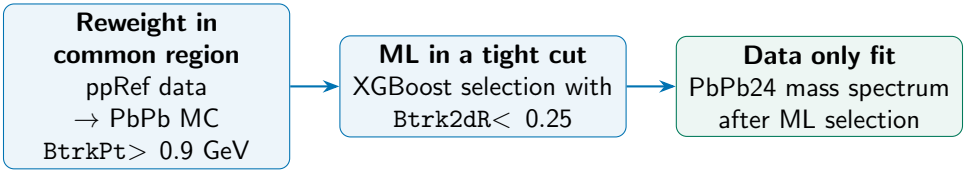








Recap: where the previous update stopped



Established

- ppRef and PbPb X MC are close in the same fiducial region.
- A X(3872) peak appears in PbPb after ML.

Still open

- Better reweighting & selection need more study.
- 2023 PbPb data could be added.

Updates from last week

$\psi(2S)$ ppRef weight $\rightarrow$ PbPb

ppRef $\psi(2S)$ sPlot-weighted data $\rightarrow$ BDT reweighting

↓

Transfer the weight to PbPb24 $\psi(2S)$ MC

↓

PbPb fit and sPlot

↓

Weighted MC vs. PbPb sPlot-weighted data

Question: can PbPb data validate the transferred weight?

Add PbPb23 to the X fit

Transfer the existing X ppRef weight to PbPb23 MC

↓

Year-specific XGBoost model and score threshold

↓

PbPb23 and PbPb24 as separate categories

↓

Simultaneous DATA-only fit

Question: does the added year materially strengthen the result?

$\psi(2S)$ reweighting, ML and fit configuration

Common region

- $10 < \texttt{Bpt} < 50$ GeV
- $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$
- $0.9 < \texttt{Btrk1Pt}, \texttt{Btrk2Pt} \leq 4.5$ GeV
- $2 < \texttt{Btktkpt} \leq 8$ GeV
- $\texttt{Btrk1dR} \leq 0.45$
- $\texttt{Btrk2dR} \leq 0.25$

Reweighting variables

- `Bcos_dtheta`
- `Btktkpt`
- `Bchi2Prob`
- `Btrk2Pt`
- `Btrk1Pt`
- `Btrk1dR`

ML training samples

- Signal: reweighted PbPb24 $\psi(2S)$ MC
- Background: PbPb24 DATA sidebands
- $3.60 < m < 3.65$ or $3.75 < m < 3.80$ GeV
- Same common region

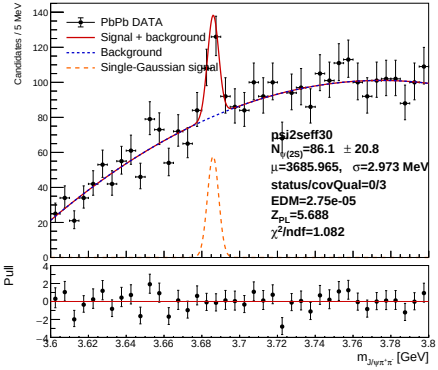
ML variables

`Btrk1dR`, `Btktkpt`, `Btrk1Pt`, `Bchi2Prob`, `Bcos_dtheta`, `Btrk2Pt`

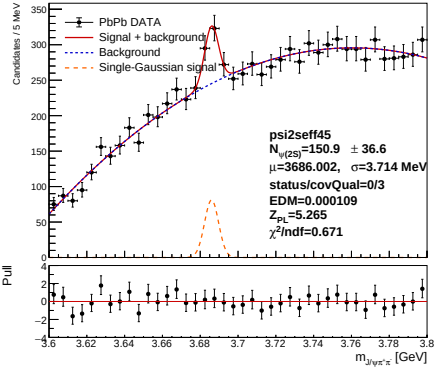
DATA-only fit model

Extended unbinned fit in $3.60 < m < 3.80$ GeV: single Gaussian signal + second-order Chebyshev background. μ starts at 3.686097 GeV and floats within ± 5 MeV; $1 \leq \sigma \leq 8$ MeV.

PbPb $\psi(2S)$ DATA-only fits



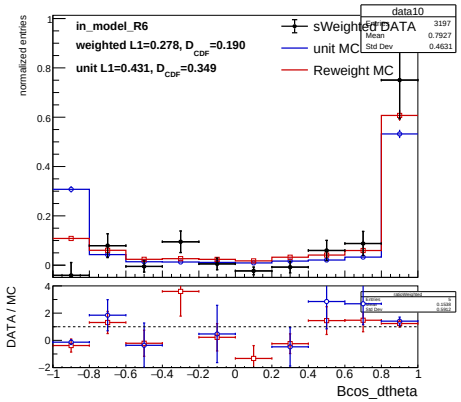
signal efficiency	30%
fitted yield	86.10 ± 20.78
mean / σ	3.68597 GeV / 2.97 MeV
status / covQual	0/3



signal efficiency	45%
fitted yield	150.90 ± 36.59
mean / σ	3.68600 GeV / 3.71 MeV
status / covQual	0/3

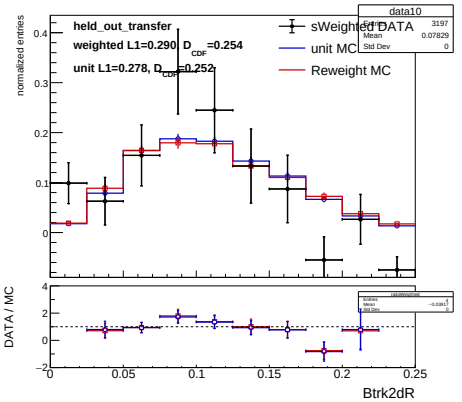
sPlot-weighted PbPb DATA vs reweighted MC

In-model example: `Bcos_dtheta`



weighted CDF distance = 0.190
used in the reweighting model

Held-out example: `Btrk2dR`



weighted CDF distance = 0.254
not used for reweighting; tests reliability on other variables

All 12 comparisons at xeff30

In-model reweighting variables

variable	before RW	after RW
Bcos_dtheta	0.349	0.190
Btktkpt	0.232	0.212
Bchi2Prob	0.129	0.175
Btrk2Pt	0.066	0.056
Btrk1Pt	0.139	0.128
Btrk1dR	0.048	0.032

Held-out transfer variables

variable	before RW	after RW
Btrk2dR	0.252	0.254
BtrkPtimb	0.144	0.147
BtktkvProb	0.159	0.130
Bpt	0.134	0.128
|By|	0.104	0.114
BQvalue	0.204	0.211

$N_{\text{eff}} = (\sum w)^2 / \sum w^2$ measures the statistical power left after signed-weight cancellation. Here $N_{\text{eff}} = 26.7$ is small, so the sPlot-weighted DATA shapes are not yet reliable enough to validate the transfer.

Why the $\psi(2S)$ validation stopped

quantity	xeff30	xeff45 one-off
score cut	0.898424	0.794011
selected DATA entries	3197	9434
DATA-only fitted yield	86.10 ± 20.78	150.90 ± 36.59
signed sPlot sum	86.013	150.996
$\sum w^2$	277.149	870.049
signed $N_{\mathrm{eff}} = (\sum w)^2 / \sum w^2$	**26.694**	**26.205**
negative-weight fraction	89.115%	88.616%
variable-comparison execution	12/12, low-N_{eff} override	0/12, gate stop

Established

RW, migration, ML, fit and sPlot interfaces are operational.

Current limitation

Loosening to xeff45 increases candidates and yield, but does not improve signed N_{eff}.

Physics validation remains statistics-limited and inconclusive.

Adding PbPb23: reweighting, ML and fit configuration

ppRef-derived reweighting

- Original: ppRef X MC
- Target: ppRef X sPlot-weighted DATA
- Same weight applied separately to PbPb23 and PbPb24 X MC

Variables

`Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk2Pt`, `Btrk1Pt`, `Btrk1dR`

RW region

$10 < \texttt{Bpt} < 50$ GeV, $|\texttt{By}| < 1.6$, $\texttt{BQvalue} < 0.15$, $\texttt{BtrkPt} > 0.9$ GeV, $2 \leq \texttt{Btktkpt} \leq 8$ GeV, $\texttt{Btrk1dR} \leq 0.45$, $\texttt{Btrk2dR} \leq 0.40$

Year-specific ML training

- Signal: reweighted X MC of the same year
- Background: DATA in $[3.82, 3.85] \cup [3.89, 3.92]$ GeV
- Separate PbPb23 and PbPb24 classifiers
- test AUC: 0.89756/0.89587

Variables

`Btrk1dR`, `Btktkpt`, `Btrk1Pt`, `Bchi2Prob`, `Bcos_dtheta`, `Btrk2Pt`

ML and fit region

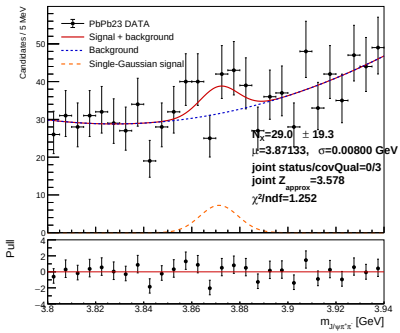
Same as above, but $\texttt{Btrk2dR} < 0.25$. Year-specific score thresholds at common weighted X efficiency.

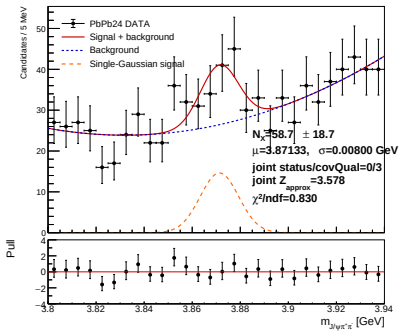
Simultaneous DATA-only fit

- PbPb23 and PbPb24 are separate categories
- Extended unbinned fit in $3.80 < m < 3.94$ GeV
- Single Gaussian signal
- Shared mean: $3.86 < \mu < 3.88$ GeV
- Independent widths: $2 \leq \sigma \leq 8$ MeV
- Independent yields and Chebyshev-2 backgrounds

Same RW–ML–fit chain for both years; only the year category and trained classifier differ.

PbPb23 and PbPb24 simultaneous fits at xeff25



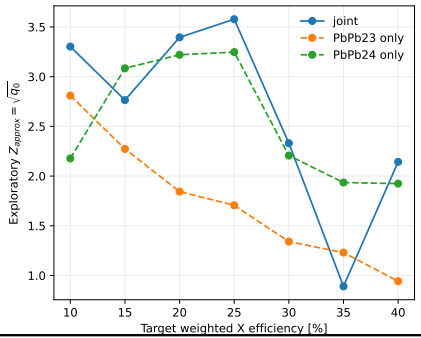


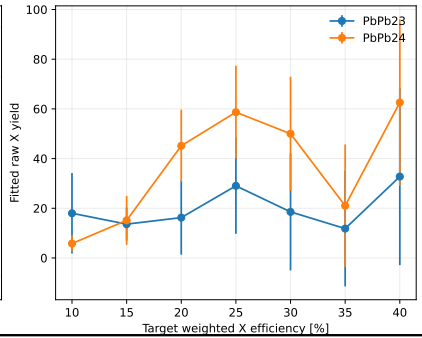
selected candidates	972
score threshold	0.952613
fitted yield	29.03 ± 19.29
Gaussian width	8.000 MeV [upper bound]

selected candidates	870
score threshold	0.953525
fitted yield	58.67 ± 18.68
Gaussian width	8.000 MeV [upper bound]

Shared fitted mean: 3.87133 GeV; fit status / covQual: 0/3.

X fit scan after adding PbPb23





target weighted X efficiency	10%	15%	20%	25%	30%	35%	40%
significance	3.303	2.764	3.396	3.578	2.331	0.891	2.143

Matched xeff25 comparison PbPb24-only: 3.247; two-year joint: 3.578; change: +0.33.

Interpretation The change is modest. Uncalibrated local likelihood proxy; both widths hit the upper bound.

Summary and Outlook

Summary

- The $\psi(2S)$ workflow is working.
- The available $\psi(2S)$ statistics limit the validation of the reweighted MC variables.
- PbPb23 data have been added to the X ppRef→PbPb analysis, with preliminary fit results.
- The PbPb23 data provide only a small increase in significance.

Outlook

- Better reweighting and selection need more study.
- A physics-approved configuration is needed before a calibrated significance study.

Thanks for listening!

Reporter: Lekai Yao

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